INCH-POUND

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31 December 1991
SUPERSEDING
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7 September 1990

MILITARY SPECIFICATION

DAIRYSHAKE POWDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dairyshake powder for preparation of dairyshakes for use by the Department of Defense as a component of operational rations and for oral liquid feeding of hospitalized patients.

1.2 Classification. The dairyshakes shall be of the following types as specified (see 6.1).

Type I - Banana
Type II - Chocolate
Type III - Eggnog
Type IV - Orange
Type V - Strawberry
Type VI - Vanilla

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8910

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
TT-C-495	-	Coating, Exterior, For Tinned Food Cans
QQ-A-1876	-	Aluminum Foil
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-C-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items: Clothing, Personal Equipment and Equipage; General Specifications For

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. Army Natick Research, Development, and Engineering Center

2-9-340 - Pouch, Dairyshake Powder

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(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Nonfat Dry Milk (Spray Dried and Instantized)
U.S. Standards for Grades of Dry Whey
General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products
DA Instruction 918-RL (Resident Grading Quality Control Service Programs and Laboratory Analysis)

(Copies are available from the Dairy Standardization Section, Dairy Division, P.O. Box 96456, Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20090-6456.)

United States Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400CD, Arlington, VA 22201-3301.)

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AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Milk, nonfat, dry, instant. Nonfat, instant, dry milk shall be U.S. extra grade and U.S. low-heat as specified in U.S. Standards for Grades of Nonfat Dry Milk (Spray Dried and Instantized).

3.2.2 Creaming agent, non-dairy. The non-dairy creaming agent shall be a cold water dispersible creaming agent. It shall possess a white to cream color and be free of hard lumps, and 100 percent shall pass through a U.S. Standard No. 20 sieve. It shall have a fat content ranging from 65 to 75 percent and a moisture content that does not exceed 5.0 percent (see 6.3.1). The non-dairy creaming agent shall be manufactured in a plant approved by the Dairy Grading Section, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, prior to submission of offer. Approved plants must satisfactorily meet the requirements contained in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products.

3.2.3 Sugar, white, powdered. Sugar shall be white refined, powdered cane or beet sugar or a combination thereof, and shall comply with MIL-STD-900.

3.2.4 Cocoa. The cocoa powder shall conform to the Standard of Identity for medium fat cocoa (21 CFR 163.113) of the Food and Drug Administration. The cocoa shall be non-dutched, and not less than 98 percent, by weight, shall pass through a U.S. Standard No. 20 sieve.

3.2.5 Dry sweet dairy whey. The dry whey shall be U.S. Extra Grade Dry Sweet Type Whey as defined in the U.S. Standards for Grades of Dry Whey.

3.2.6 Starch, granular, waxy-maize, unmodified, instant. The starch shall be a pure, off-white, odorless, granular powder. It shall be unmodified and instantized. It shall have a viscosity range of 800 to 1000 cps per minute using a Brookfield viscometer. The starch shall have a maximum moisture of 9.0 percent and a pH of 4.0 to 6.5. It shall display high clarity, sheen, and hot/cold storage stability. The starch shall be bland with no objectionable off-notes and shall comply with MIL-STD-900 (see 6.3.2).

3.2.7 Calcium caseinate. The calcium caseinate shall be a spray dried, totally reacted, flavor-reduced milk protein manufactured from selected, premium quality, edible casein. It shall possess a white to pale cream

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color, and a bland and clean odor and flavor. The calcium caseinate shall exhibit high opacity in water suspensions, and a moderate water binding capacity. The protein content (on a dry basis) shall range from 80 to 90 percent; the moisture content shall range from 6.5 to 7.5 percent; the fat content shall range from 1.0 to 1.3 percent; and the percent lactose shall not be greater than 0.2 percent (see 6.3.3).

3.2.8 Flavoring, banana, artificial. Artificial banana flavoring shall be powdered and shall have an odor and flavor typical of banana (see 6.3.4).

3.2.9 Color, Red No. 40. Red No. 40 color shall be in powdered form, and shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199).

3.2.10 Color, Yellow No. 5. Yellow No. 5 color shall be in powdered form, and shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199).

3.2.11 Flavoring, vanilla, artificial. Artificial vanilla flavoring shall be powdered and shall have an odor and flavor typical of vanilla (see 6.3.5).

3.2.12 Flavoring, french vanilla, natural and artificial. Natural and artificial french vanilla flavoring shall be powdered and shall have an odor and flavor typical of french vanilla (see 6.3.6).

3.2.13 Flavoring, eggnog, artificial. Artificial eggnog flavoring shall be powdered and shall have an odor and flavor typical of eggnog (see 6.3.7).

3.2.14 Titanium dioxide. Titanium dioxide shall be water dispersible and shall comply with the Food Chemicals Codex.

3.2.15 Flavoring, chocolate, artificial. Artificial chocolate flavoring shall be powdered and shall have an odor and flavor typical of chocolate (see 6.3.8).

3.2.16 Color, Yellow No. 6. Yellow No. 6 color shall be in powdered form, and shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199).

3.2.17 Flavoring, orange, artificial. Artificial orange flavoring shall be powdered and shall have an odor and flavor typical of orange (see 6.3.9).

3.2.18 Flavoring, strawberry, artificial. Artificial strawberry flavoring shall be powdered and shall have an odor and flavor typical of strawberry (see 6.3.10).

3.3 Preparation and processing.

3.3.1 Product formulation. The formulas for the six dairyshakes shall be as follows:

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3.3.1.1 Formulation for banana dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	45.135
Non-dairy creaming agent	27.305
Sugar, white, powdered	18.280
Dry sweet dairy whey	4.540
Starch, instant	2.720
Calcium caseinate	1.810
Flavoring, banana, artificial	0.200
Color, Yellow No. 5	0.010

3.3.1.2 Formulation for chocolate dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	39.000
Non-dairy creaming agent	23.440
Sugar, white, powdered	23.400
Cocoa	6.200
Dry sweet dairy whey	3.900
Starch, instant	2.300
Calcium caseinate	1.600
Flavoring, chocolate, artificial	0.120
Flavoring, vanilla, artificial	0.020
Flavoring, french vanilla, natural and artificial	0.020

3.3.1.3 Formulation for eggnog dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	45.330
Non-dairy creaming agent	27.300
Sugar, white, powdered	18.100
Dry sweet dairy whey	4.500
Starch, instant	2.700
Calcium caseinate	1.800
Flavoring, eggnog, artificial	0.120
Titanium dioxide	0.085
Flavoring, vanilla, artificial	0.030
Flavoring, french vanilla, natural and artificial	0.030
Color, Yellow No. 5	0.004
Color, Yellow No. 6	0.001

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3.3.1.4 Formulation for orange dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	45.200
Non-dairy creaming agent	27.232
Sugar, white, powdered	18.100
Dry sweet dairy whey	4.500
Starch, instant	2.700
Calcium caseinate	1.800
Flavoring, orange, artificial	0.450
Color, Yellow No. 5	0.013
Color, Red No. 40	0.005

3.3.1.5 Formulation for strawberry dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	45.300
Non-dairy creaming agent	27.301
Sugar, white, powdered	18.100
Dry sweet dairy whey	4.500
Starch, instant	2.700
Calcium caseinate	1.800
Flavoring, strawberry, artificial	0.230
Flavoring, vanilla, artificial	0.030
Flavoring, french vanilla, natural and artificial	0.030
Color, Red No. 40	0.009

3.3.1.6 Formulation for vanilla dairyshake.

Ingredients	Percent by weight
Milk, nonfat, dry, instant	45.3998
Non-dairy creaming agent	27.2000
Sugar, white, powdered	18.2000
Dry sweet dairy whey	4.5000
Starch, instant	2.7000
Calcium caseinate	1.9000
Titanium dioxide	0.0379
Flavoring, vanilla, artificial	0.0300
Flavoring, french vanilla, natural and artificial	0.0300
Color, Yellow No. 5	0.0018
Color, Red No. 40	0.0005

3.3.2 Mixing of dry ingredients. The dairyshake ingredients shall be uniformly blended in a suitable mixer until no lumps exist and the product is homogeneous.

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3.3.2.1 Instantization. The homogeneously mixed product shall be commercially instantized in such a manner as to ensure compliance with the finished product requirement.

3.3.3 Pouch filling and sealing. After mixing, the product shall be completely packaged and sealed in pouches conforming to the requirements in 5.1.1.1. Each pouch shall be filled to comply with the finished product requirements.

3.3.4 Can filling and sealing. After mixing, the product shall be filled into cans conforming to the requirements of 5.1.1.1.2. Each can shall be filled to comply with the finished product requirements.

3.4 Finished product requirements.

3.4.1 Dry product. The dry product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Contents of each individual pouch shall weigh not less than 95 grams and not more than 105 grams.

e. The lot average net weight of the pouch contents shall be not less than 100 grams.

f. The product shall be free from caking and lumps which do not fall apart under light pressure.

g. Not more than 2.5 grams of product from a pouch or more than 12.5 grams of product from a can shall remain when the product is passed through a U.S. Standard No. 7 sieve.

h. The moisture content shall not exceed 3.5 percent.

i. The aerobic plate count shall be not greater than 50,000 per gram in four of five samples and not greater than 75,000 in any individual sample.

j. The E. coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any individual sample. The Salmonella test shall be negative.

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k. The net weight of each 401 x 602 can shall be not less than 475 grams and not more than 525 grams.

l. The average net weight of the 401 x 602 can contents shall be not less than 500 grams.

3.4.2 Hydrated product. Hydration of the dry powder shall be in accordance with the specified preparation instructions on the pouch or can label (as applicable). The rehydrated product shall comply with the following requirements:

a. After hydration, no particles larger than 0.5 mm shall remain when the product has passed through a U.S. Standard No. 20 sieve.

b. The hydrated product shall have a flavor and odor that is characteristic of the specified dairyshake.

c. The hydrated product shall have a color as specified:

banana	-	pale to medium yellow
chocolate	-	light to medium brown
eggnog	-	medium to dark cream
orange	-	pale to medium orange
strawberry	-	light to medium pink
vanilla	-	pale to medium cream

d. The hydrated product shall have a consistency that allows it to be easily sipped through a straw.

3.4.3 Palatability. The hydrated product shall be equal to or better than the approved preproduction sample in palatability and overall appearance (see 6.1).

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packing or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency (USDA, AMS, Dairy Division).

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1. Contractor's responsibility. Inspection and acceptance by USDA, Agriculture Marketing Service (AMS), Dairy Division, shall not relieve the contractor of obligation and responsibility to deliver a product complying

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with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA, AMS, Dairy Division, for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA, AMS, Dairy Division. The USDA, AMS, Dairy Division, will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Pouch material examination. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

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Material requirement	Requirement paragraph	Test procedure
Thickness of films	5.1.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.1	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.1	Laboratory evaluation
Color of laminated material	5.1.1.1	Visual examination

4.5.1.3 Unfilled pouch certification. A certificate of compliance may be accepted as evidence that unfilled pouches conform to the requirements specified in 5.1.1.1.1. When deemed necessary by the USDA, AMS, Dairy Division, testing of unfilled preformed pouches for seal strength shall be as specified in 4.5.1.3.1.

4.5.1.3.1 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample unit shall be one unfilled pouch. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the average seal strength requirement specified in 5.1.1.1.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.4 Unfilled can inspection. Conformance of unfilled cans to the requirements specified in 5.1.1 shall be determined by examination of certificates of compliance or of other valid documents. Any nonconformance shall be cause for rejection of the can lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual

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examination or by review of records of formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Net weight inspection.

4.5.3.1 Net weight inspection of pouches. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 95 grams or more than 105 grams shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram. In addition, the lot shall be rejected if the sample average net weight is less than 100 grams.

4.5.3.2 Net weight inspection of cans. The net weight of the filled and sealed cans shall be determined by weighing each sample unit on a suitable scale tared with a representative empty can and lid. Any individual net weight less than 475 grams or greater than 525 grams shall be classified as a minor defect. The results shall be reported to the nearest gram. The lot shall be expressed in cans. The sample unit shall be one can. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the average net weight is less than 500 grams.

4.5.4 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE I. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Pouch has foreign odor
102		Not heat sealed as specified
103		Not clean 2/
104		Tear, hole, or open seal
105		Evidence of delamination 3/
106		Seal width not as specified

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TABLE I. Filled and sealed pouch defects 1/ (cont'd)

Category		Defect
Major	Minor	
107		Closure seal not located above pull tab or located less than 1/8 inch above tear notch
108		Presence of foldover wrinkles, occluded matter, and entrapped moisture or grease that reduces the effective closure seal to less than 1/16 inch 4/
109		Product nomenclature or directions for use missing, incorrect, or illegible
110		Label torn or otherwise damaged so as to obliterate any of the markings on the label
111		Zipper reclosure feature missing
	201	Thickness of pouch exceeds 1/2 inch
	202	Pull tab or tear notch missing or not as specified
	203	Pull tab or tear notch not located as specified
	204	Label not properly affixed to pouch (for example, label raised or peeling back from edges of corners)
	205	Nomenclature letters not required size
	206	Marking smear 5/
	207	Zipper reclosure feature not located as specified
	208	Evidence of delamination 3/

1/ Any evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass metal filings, etc.) or generally detracts from the clean appearance of the pouch. The following examples shall not be scored as defects for unclean:

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a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the pouch or by gently brushing the pouch with a clean, dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

3/ Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed 10 times by rotating both hands in alternating clockwise-counter-clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be scored as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the delaminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (plus or minus 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be scored as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as minor.

4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide from side seal to side seal that produces a hermetically sealed pouch.

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5/ The marking on the pouch label shall not smear when tested by rubbing with moderate pressure with a dry finger after allowing markings to dry at least 12 hours.

4.5.5 Pouch vacuum examination. When pouches are sealed in accordance with partial vacuum requirements specified in 5.1.1.1.1, and they shall be subjected to a test for residual gas volume as specified in 4.5.5.1 or 4.5.5.2, not less than 96 hours after filling and sealing. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2. Any test failure shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.5.1 Residual gas volume destructive test. Volume of residual gases shall be determined as follows: The samples for testing shall be opened under $75^{o} \pm 5^{o}$F water and the gases shall be collected by water displacement in a graduated cylinder or other calibrated tube. The volume of the gases shall be reported to the nearest 0.1 cubic centimeter. Any residual gas volume that does not fall within the range of 80 to 125 cubic centimeters shall be considered a test failure.

4.5.5.2 Residual gas volume non-destructive test. A Meade tester or similar modified vacuum dessicator type apparatus shall be used to test the pouches. Individual samples, suspended from a beam-type balance shall be weighed in the water (completely submerged) of the test apparatus to the nearest 0.1 gram. The temperature of the water during the test shall be $75^{o} \pm 5^{o}$ F. If the sample does not sink, a small arbitrary weight shall be attached to the sample. After weighing, the individual samples (weight attached if used) shall be detached from the balance and resubmerged in the water of the tester. With the lid in place, the sample shall be made to float by reducing the air pressure over the water by means of a vacuum pump. The reduced pressure, in millimeters of mercury, at which the sample just floats below the water surface (still completely submerged), is recorded as the pressure at neutral buoyancy. When making the above determinations, care should be exercised to prevent small bubbles of air from adhering to the sample. The atomospheric pressure, in millimeters of mercury, shall be measured with a barometer. The data from the three determinations shall be used to calculate the volume of residual gases to the nearest 0.1 cubic centimeter as follows:

$$\text{Volume of residual gases } (cm^3) = \frac{\text{Pressure at neutral buoyancy} \times \text{weight in water (g)}}{\text{ATM pressure} - \text{Pressure at neutral buoyancy}}$$

Any residual gas volume that does not fall within the range of 80 to 125 cubic centimeters shall be considered a test failure.

4.5.6 Pouch closure seal strength testing. The filled and sealed pouches shall be tested for closure seal strength in accordance with ASTM F 88, except that the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or

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lower clamp may be used. Test specimens shall be cut 1/2 inch in width and to a length suitable for proper mounting. Three adjacent specimens shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.1 shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-1.

4.5.7 Product examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in pouches or cans. The sample unit shall be one filled and sealed pouch or can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	Applicable to dry product
101	Product is not free from caking or lumps which do not fall apart under light pressure
102	More than 2.5 grams of product from a pouch or more than 12.5 grams of product from a can remains on a U.S. Standard No. 7 sieve 3/
	Applicable to hydrated product 4/ 5/
103	Particles larger than 0.5 mm remain after product is passed through a U.S. Standard No. 20 sieve
104	The hydrated product does not have flavor or odor characteristic of the specified dairyshake
105	The hydrated product color is not as specified: banana - pale to medium yellow chocolate - light to medium brown eggnog - medium to dark cream orange - pale to medium orange strawberry - light to medium pink vanilla - pale to medium cream
106	The product does not have a consistency which allows it to be easily sipped through a straw

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1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.4.3).

3/ The powder shall be poured onto a U.S. Standard No. 7 sieve and vigorously shaken.

4/ Hydration of the dry powder from pouches shall be accomplished by mixing each pouch with 6 ounces of cold water in a high speed blender for 30 seconds.

5/ Hydration of dry powder from the 401 x 602 can shall be accomplished by mixing 100 grams with 6 ounces of cold water in a high speed blender for 30 seconds.

4.5.8 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture in Dried Fruits Methods (vacuum drying) in the Fruit and Fruit Product Chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC) with the exception that the product shall be dried for 16 hours. Test results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1h shall be classified as a major defect and shall be cause for rejection of the lot. The lot size shall be expressed in pouches or cans. The sample unit shall be one filled and sealed pouch or can. The inspection level shall be S-2.

4.5.9 Microbiological testing. Five filled and sealed pouches or cans shall be selected at random from the lot regardless of lot size. The pouches or cans shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the AOAC; Chapter: Microbiological Methods; Section: Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before blending the sample. For product packaged in 401 x 602 cans, mix 6 ounces cold water with 100 grams of powder and allow to stand for 15 minutes before blending the sample. Continue with the procedure as directed in the Official First Action. Salmonella testing shall be in accordance with DA Instruction 918-RL, Instructions for Resident Grading Quality Control Service Programs and Laboratory Analysis, USDA, AMS Dairy Division, Dairy Grading Section. Any result not conforming to the microbiological requirements in 3.4.1i and 3.4.1j shall be classified as a major defect and shall be cause for rejection of the lot.

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4.5.10 Vacuum testing (applicable to cans only). The filled and sealed cans shall be tested for vacuum in accordance with the following procedures. The vacuum readings shall be taken with a puncture-type vacuum gauge, making the puncture as near as possible to the double seam to minimize error due to distortion of the end. Vacuum shall mean that there is a perceptible movement of the indicator needle of the vacuum gauge. For normal inspection purposes, a correction of 1 inch of mercury vacuum shall be added to the gauge reading for each 1000 feet above sea level at which the determination is made. If greater accuracy is considered necessary, sample cans shall be submitted to a laboratory for vacuum determination corrected to standard conditions (68 degrees F and 29.9 inches of mercury). Correction for gauge and headspace errors shall be included. Results shall be reported to the nearest inch. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. Sample cans selected shall be free from dents or other visible damage. The inspection level shall be S-2. Any nonconformance to the requirement in 5.1.1.1.2 shall be classified as a major defect and shall be cause for rejection of the lot. The contents of these cans may be used for the product inspection in 4.5.7.

4.5.11 Oxygen in headspace testing (applicable to cans only). The determination of the oxygen content in the headspace gas shall be by using an electronic oxygen analyzer which operates on the principle of the difference in partial pressure of oxygen between the oxygen reference and the oxygen content of the sample as detected by a porous ceramic Zirconica sensor, such as the Illinois Instruments analyzer or its equivalent; or on the principle of paramagnetic resonance such as the Servomex analyzer, or its equivalent. The oxygen analyzer shall be calibrated to a known standard prior to testing the headspace gas of the product. Product shall be tested within 24 hours of packing. Results shall be reported to the nearest 0.1 percent. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-2. Any nonconformance to the requirement in 3.4 shall be classified as a major defect and shall be cause for rejection of the lot. The contents of these cans may be used for the product inspection in 4.5.7.

4.5.12 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4.2). In addition, the following defect shall be classified as follows:

Minor: Reusable plastic lid missing, ripped, torn, or cut.

4.5.13 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.

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Minor: Other required markings missing, incorrect, or illegible.
Not a snug-fitting container.
Number of pouches or cans not as specified.

4.5.14 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packing.

5.1.1.1.1 Pouches. Each dairyshake shall be unit packed in a pouch conforming to figure 1. The pouch shall be fitted with a reclosable zipper feature, minimum 1/8 inch wide, located as specified in figure 1. As an alternative to "V," "C," or "U" shaped tear notch, a pull tab line may be used. The inside dimensions of the pouch shall be 4-3/4 by 9-1/4 (± 1/8) inches. Side and bottom seals shall be 3/8 (± 1/8) inch wide. The end of the preformed pouch may be provided with a 1/4 ± 1/8 inch extended lip to facilitate opening and filling. The pouch material shall be a 3-ply laminate consisting of a 0.0020-inch thick ionomer or polyethylene film laminated or extrusion-coated to 0.00035-inch thick aluminum foil, which is then laminated to 0.00050-inch thick polyester. The three layers shall be laminated so that the aluminum foil is between the other two layers. The pouch shall be formed with the polyester on the exterior. The pouch material shall be approved for food use and shall show no evidence of delamination or degradation when heat sealed or fabricated into pouches. The complete exterior surface of the pouch shall be uniformly colored in the range of Green 34079 through 34087 or 24052 through 24087, or Brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. The pouches shall be filled in accordance with the net weight requirements of 3.4.1. The filling operation shall be carefully executed to prevent contamination of the closure seal. The filled pouch shall be closed by evacuating the air within a sealing chamber under a vacuum of not less than 28 inches of mercury, then by flushing back with nitrogen gas to atmospheric pressure before sealing. Alternatively, the filled pouch may be closed under a partial vacuum of not less than 8 and not more than 12 inches of mercury and shall have a residual gas volume of 80 to 125 cubic centimeters when tested as specified in 4.5.5. The pouch shall be closed with a continuous heat seal closure not less than 1/4 inch wide extending across the complete width of the pouch. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The closure seal location shall be as shown in figure 1. Both the pouch manufacturer's heat seals and the closure

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heat seal shall have an average seal strength of 6 pounds per inch of width, and each individual test specimen shall have a seal strength of not less than 5 pounds per inch width when tested in accordance with 4.5.1.3 and 4.5.6. The product shall be distributed within the sealed pouch to attain a thickness of not more than 1/2 inch.

5.1.1.1.2 Cans. Five hundred grams (see 3.3.4) of dairyshake shall be filled into an open-top style, round metal can, size 401 x 602, with welded side seam and compound-lined, double seamed ends. The can shall be made throughout from not less than 0.25 pound per base box electrolytic tin plate. The base plate of the can shall be of sufficient temper to protect the product during shipment and storage. The can shall be coated on the outside with a coating conforming to type I or when specified (see 6.1) type III of TT-C-495. The can shall be hermetically sealed under a vacuum of not less than 15 inches of mercury when tested in accordance with 4.5.10. Alternatively, the can may be sealed under an atmosphere of nitrogen so that the oxygen content of the gases in the filled and sealed can do not exceed 3.0 percent when tested in accordance with 4.5.11. When cans are closed under vacuum, the oxygen requirement is not applicable.

5.1.1.1.2.1 Plastic lid. The filled and sealed cans shall have a close fitting plastic lid, size 401, affixed securely to one end of the can. The plastic lid shall be made of reusable, self-sealing polyethylene. The lid shall be of a high pressure type, snap on, pull off flexible basis.

5.1.2 Level C.

5.1.2.1 Pouches. One hundred grams (see 3.4.1) of product shall be filled into pouches as specified in 5.1.1.1.1.

5.1.2.2 Cans. Five hundred grams (see 3.4.1) of dairyshake shall be filled into cans as specified in 5.1.1.1.2, except that cans with or without exterior commercial coating will be acceptable. Alternatively, cans may be made from 0.20 pound per base box electrolytic tin plate provided the cans have exterior commercial coating. A plastic reclosure lid shall be affixed to one end of the can.

5.2 Packing. Packing shall be level B or Commercial as specified (see 6.2).

5.2.1 Level B packing.

5.2.1.1 Pouches. Twenty-four pouches of dairyshakes of one flavor only, preserved as specified in 5.1, shall be packed flat in two stacks of twelve pouches, in a snug-fitting fiberboard shipping container conforming to style RSC, grade V3c, V3s, or V4s of PPP-B-636. The approximate inside dimensions of the shipping container shall be 11 inches long, 9-1/2 inches wide, and 6-1/2 inches deep. Approximate dimensions are furnished as a guide only.

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Shipping containers shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.1.2 Cans. Twenty-four cans of dairyshakes of one flavor only, preserved as specified in 5.1, shall be packed on end, four in length, three in width, and two in depth in a snug-fitting fiberboard shipping container as specified in 5.2.1.1. Tiers of cans shall be separated by a fiberboard pad.

5.2.2 Level C packing. Dairyshakes of one flavor only, preserved in pouches as specified in 5.1, or twenty-four cans of dairyshakes of one flavor only, preserved as specified in 5.1, arranged on end, four in length, three in width and two in depth, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable, except fiberboard containers shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Unit loading. When specified (see 6.2), the product, packed as specified in 5.2.1 or 5.2.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling.

5.4.1 Pouches. Each pouch shall have labeling that complies with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder. The label shall be pressure-sensitive and shall be colored a dull, non-reflective color other than white. All printing thereon shall be a black, legible type with indelible ink. The label shall be applied to one face of the empty pouch with sufficient pressure to ensure its adherence to the pouch. Labels shall contain the following information:

a. Each flavor of dairyshake shall have a separate label (block letters not less than 5/32 inch high):

   1. For the banana dairyshake, the label shall read:
      DAIRYSHAKE POWDER, ARTIFICIAL BANANA FLAVOR

   2. For the chocolate dairyshake, the label shall read:
      DAIRYSHAKE POWDER, ARTIFICAL CHOCOLATE FLAVOR

   3. For the eggnog dairyshake, the label shall read:
      DAIRYSHAKE POWDER, ARTIFICIAL EGGNOG FLAVOR

   4. For the orange dairyshake, the label shall read:
      DAIRYSHAKE POWDER, ARTIFICIAL ORANGE FLAVOR

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5. For the strawberry dairyshake, the label shall read:
DAIRYSHAKE POWDER, ARTIFICIAL STRAWBERRY FLAVOR

6. For the vanilla dairyshake, the label shall read:
DAIRYSHAKE POWDER, ARTIFICIAL VANILLA FLAVOR

b. For each flavor, pouches shall be labeled with the list of ingredients which shall be in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

c. For each flavor, pouches shall be labeled with the following nutritional information:

Protein: 18g (minimum)
Fat: 19g (maximum)
Carbohydrates: 55g (maximum)
Calories: 463 Kcal (minimum)

d. For each flavor, pouches shall be labeled with the following mixing instructions:

(1) For oral liquid feeding of hospital patients:

Empty contents of pouch into a high-speed food blender.
Add 6 ounces cold water or milk.
Blend for 30 seconds.
Serve within 4 hours or refrigerate.

(2) For field use as an operational ration component:

Open pouch.
Add 1/4 canteen cup (6 ounces) cold water.
Close pouch.
Shake vigorously for 30-60 seconds.
Consume promptly.

e. For each flavor, pouches shall be labeled with the following net weight:

3.5 ounces (100 grams net).

f. Contractor's name and address shall be on each pouch label.

g. Date of pack 1/

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1/ Each pouch shall have the date of pack noted by using a four digit code beginning with the final digit of the current year and followed by the three digit Julian day code. (For example, March 19, 1991 would read 1079.) The Julian day code shall represent the date the product is unit packed into the pouch. When a non-contact type marker is used or when a soft rubber contact type marker is used, the code may be located anywhere on the pouch, provided the pouch material is not damaged and the code is not superimposed over any other printing. Embossed codes are acceptable provided they are located in the heat seal other then the closure seal a minimum of 1/16 inch away from the inside edge of the heat seal.

5.4.2 Cans. Cans shall be labeled in accordance with MIL-L-1497, except that the body of the can shall be labeled. Labeling information shall be as follows:

a. Dairyshake flavor (see 5.4.1a)

b. INGREDIENTS: (to be listed in accordance with Federal Food Drug and Cosmetic Act and regulations promulgated thereunder)

c. DIRECTIONS FOR USE:

Place 1 cup dry measure (100 grams) of product in a high speed food blender.
Add 6 ounces cold water.
Blend for 30 seconds.
Serve within 4 hours or refrigerate.

d. Net weight: 17.5 ounces (500 grams)

e. Nutritional Information (per serving):

Servings per container:	5
Protein:	18 grams (minimum)
Fat:	19 grams (maximum)
Carbohydrate:	55 grams (maximum)
Calories:	463 Kcal (minimum)

f. Contractor name and address

g. Date of pack 1/

h. Gas packaged (if applicable)

1/ Each can shall have the date of pack noted by using a four digit code beginning with the final digit of the current year and followed by the three digit Julian day code. (For example, March 19, 1991 would read 1079.) The Julian day code shall represent the date the product is unit packed into the can.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type (see 1.2).
c. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.4.3 and 6.2).
f. Level of preservation and packing required (see 5.1 and 5.2).
g. Type and class of unit load when unit loading is required (see 5.3).
h. When type III exterior coating of cans is required (see 5.1.1.1.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements of selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Non-dairy creaming agent. It has been found that San-a-creme CWS 881 manufactured by Beatreme Food Ingredients meets the requirements of 3.2.2 and performs satisfactorily in this product.

6.3.2 Starch, granular, waxy maize, unmodified, instant. It has been found that Stir and Sperse manufactured by A.E. Staley Mfg., Division of Staley Continental, Inc. meets the requirements of 3.2.6 and performs satisfactorily in this product.

6.3.3 Calcuim caseinate. It has been found that Alanate 311 manufactured by New Zealand Milk Products meets the requirements of 3.2.7 and performs satisfactorily in this product.

6.3.4 Flavoring, banana artificial. It has been found that artificial banana 41702T manufactured by F&C International meets the requirements of 3.2.8 and performs satisfactorily in this product.

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6.3.5 Flavoring, vanilla, artificial. It has been found that artificial vanilla 650204U manufactured by Hercules meets the requirements of 3.2.11 and performs satisfactorily in this product.

6.3.6 Flavoring, french vanilla, natural and artificial. It has been found that natural and artificial french vanilla 610364U manufactured by Hercules meets the requirements of 3.2.12 and performs satisfactorily in this product.

6.3.7 Flavoring, eggnog, artificial. It has been found that artificial eggnog 23242T manufactured by F&C International meets the requirements of 3.2.13 and performs satisfactorily in this product.

6.3.8 Flavoring, chocolate, artificial. It has been found that artificial chocolate manufactured by F&C International meets the requirements of 3.2.15 and performs satisfactorily in this product.

6.3.9 Flavoring, orange, artificial. It has been found that artificial orange 25592T manufactured by F&C International meets the requirements of 3.2.17 and performs satisfactorily in this product.

6.3.10 Flavoring, strawberry, artificial. It has been found that artificial strawberry 41428T manufactured by F&C International meets the requirements of 3.2.18 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Combat field feeding
Hospital
Jaw injury
Liquid diet
Operational ration
Oral liquid feeding
Pouch food
Shelf stable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
Air Force - 50
DP - SS

Preparing activity:

Army - GL

(Project 8910-0458)

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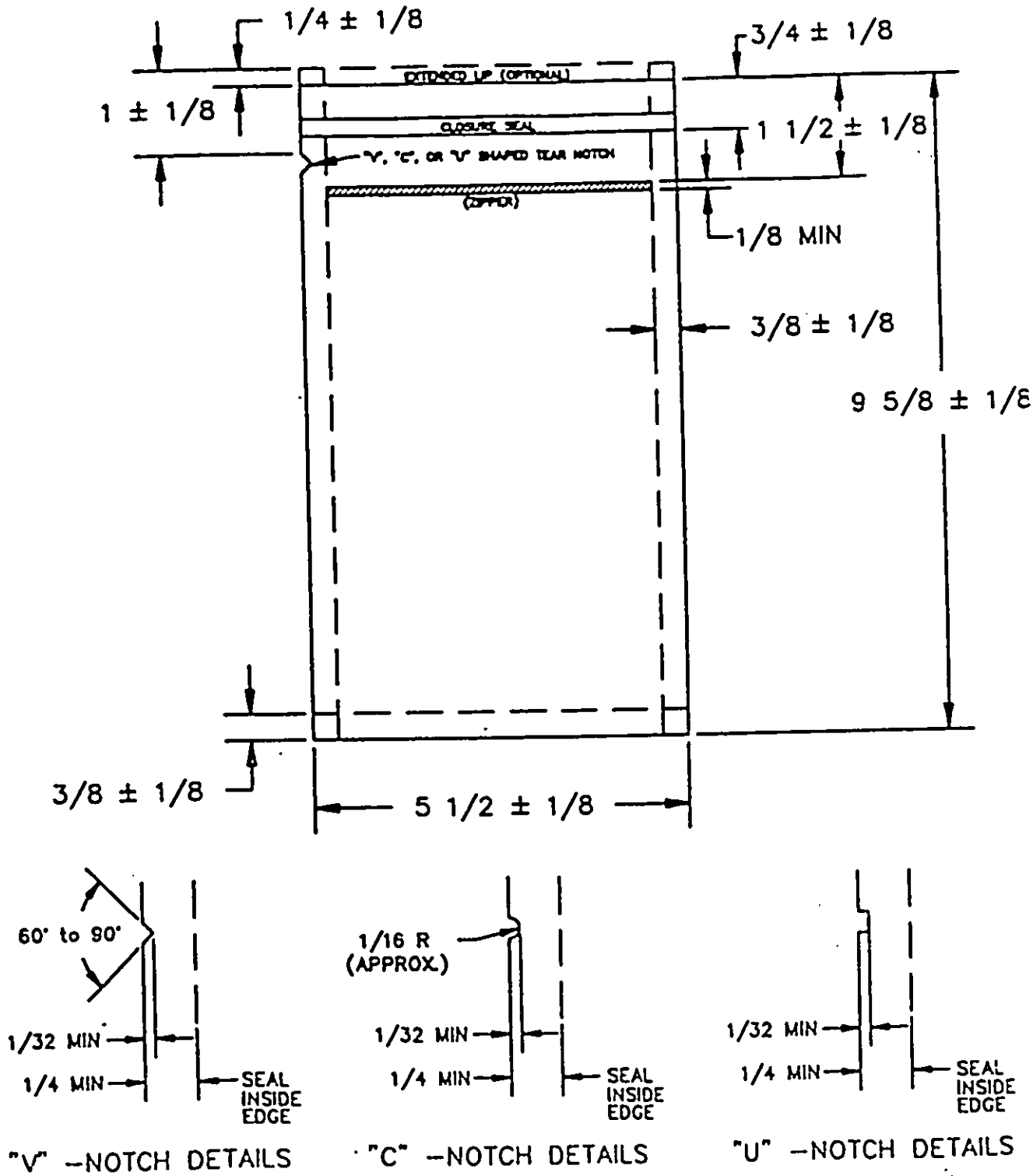


Figure 1. The pouch.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-44414A	2. DOCUMENT DATE (YYMMDD) 1991 December 31

3. DOCUMENT TITLE
Dairyshake Powder

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: 508-651-4501 (2) AUTOVON: 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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